

Submission Instructions

Read the instructions **carefully**.

- Each person submits their own theoretical part. The theoretical part should be a single file in PDF format only (no `.docx` or `.jpg`) named `ex1t_ID.pdf` (where ID is your ID).
- If you are submitting handwritten answers, make sure they are crystal clear.
- Only **one** person from each group should submit the practical part. Group should be up to two people.
- The person who submits should submit a single ZIP file named `ex1p_ID.zip` (where ID is the ID of the person submitting). The zip should contain:
 1. A folder named `code/`
 2. A folder named `output/`
 3. A `group.txt` file containing names and IDs of the team.
- Points may be reduced for submissions that fail to comply.
- Make sure you follow the News forum and HW forum for any updates.

⇒ For each problem, add **a few short your sentences under the title “Discussion”**. These can be some small insight you came up with while working on the problem or a follow-up question you have about the topic (no need to answer it).

Theoretical Part (20 pts)

Problem 1 (Communities).

- (a) When would you use clustering and when would you use community detection?
- (b) In clique percolation, which one(s), if any, of the following can result in two nodes that used to belong to a community splitting into two different communities? Justify your answer.
- (i) Adding edges to a graph
 - (ii) Removing edges from the graph
 - (iii) Increasing k
 - (iv) Decreasing k
- (c) Consider the graph defined by the following adjacencies:

A	B, C
B	A, D, E
C	A, F, I
D	B, G, E
E	B, D, H
F	C, I
G	D, H
H	E, G
I	C, F

- (1) Intuitively, which edges have the highest betweenness? Why?
- (2) Follow the algorithm we showed in class (manually) to compute the betweenness of all edges for paths starting from node B

Practical Part (80 pts)

Problem 2 (Viz (30 pts)). In this problem, you will need to **read** the JSON file from HW1 and analyze and visualize the data. Code and output should be in **code/problem2** and **output/problem2**.

- (a) Find the min, max, mean and std of each numerical variable (not nominal or ordinal).
- (b) If you chose Project Gutenberg:
 - (1) Visualize the distribution of number of subjects using a histogram.
 - (2) Use bar plots to display the top 5 years with the highest number of books, along with the number of books each of them has in your dataset.

If you chose US Presidents:

- (1) Create a new column called "Total Presidential Years", which indicates the total number of years each president was in office (sum up the time from all their terms using exact dates, round down the result to the nearest year). Plot a histogram of this column.
- (2) Create a bar plot visualizing how many presidents came from each political party. Do not count the same person more than once per party, even if he served multiple terms. Sort from highest to lowest.
- (3) Create two bar plots: one showing the top 5 presidents with the longest terms, and another showing the top 5 with the shortest terms. Sort from high to low.
- (4) Create one additional plot that reveals an interesting insight or fact about the data. Include a brief caption explaining what the plot shows.

Problem 3 (Text Mining (Coding Question)). (50 pts)

Download a movie script (e.g., from here <https://huggingface.co/datasets/IsmaelMousa/movies> but you could pick a different source), preferably one that you know. You should submit two things: The code and the answers. You can use existing NLP and viz packages (I recommend NLTK or SpaCy for NLP). For the tag clouds, you can also use online tools.

- (a) Which movie? :)
- (b) Tokenize the text. Count occurrences for each token. Plot the results (y axis: log of frequency, x axis: log rank). Also print a list of the top 20 tokens (separate from the plot). (10pt)
- (c) Repeat (b) after removing stopwords. (5pt)
- (d) Repeat (b) with stemmed text. (5pt)
- (e) Run POS-tagging on the original text. Extract all the *adj+noun phrases*. For this exercise, we will define it as one or more adjectives, followed by one or more nouns, including proper nouns (the longest such sequence). For example – delicious peanut butter cookies, oval office. Repeat (b) using adj+noun phrases as tokens. (14pt)
- (f) Show one example sentence where POS tagging made a mistake (explain). (6pt)
- (g) Create a Tag cloud (word cloud) of proper nouns (NNP, NNPS). Does it correspond to what you know about the movie? (7pt)
- (h) Write a regular expression to find the set of all strings with two consecutive repeated words, with potential punctuation between them (e.g., "well well", "so so", and "no, no"). (Hint: `\b` is a word boundary, and `\1` references the first captured match). Run on your text and report any found. (8pt)